



Barker College

Student Name:

Teacher's Initials:

2017
YEAR 11
EXAMINATION
TERM 3

Mathematics Extension 1

Staff Involved:

- AXD
- WMD
- HG
- RMH
- KJL
- ESP*

PM WEDNESDAY 13TH SEPTEMBER

110 copies

- Attempt Questions 1 to 6
- Answer in the booklets provided.
- Start each question in a new booklet.
- Write using blue or black pen for working.
- Show all necessary working. Marks may not be awarded for careless or poorly arranged working.
- Write your name and teacher's initials on each booklet.
- Board-approved calculators may be used.

Total marks – 75

Working time – 90 minutes

Topics Covered:

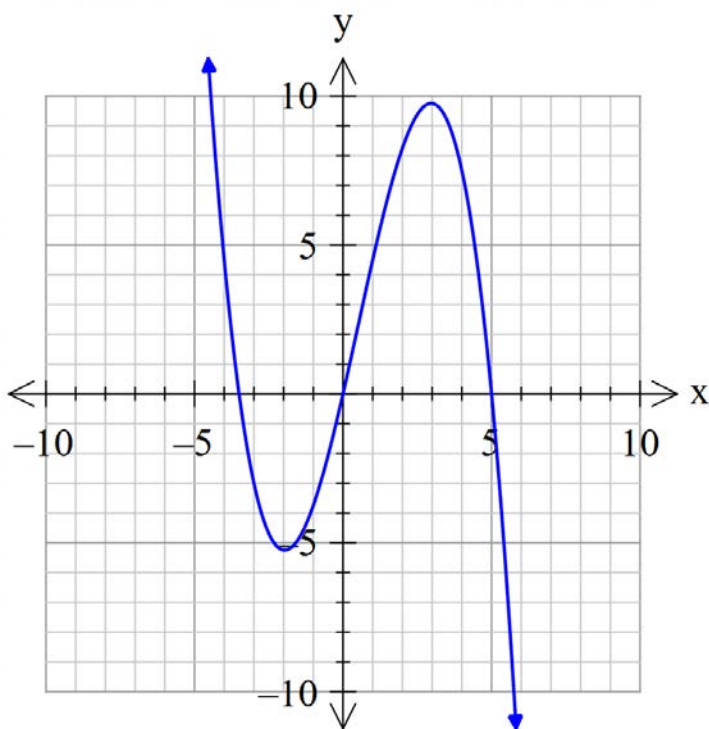
- All topics in 2 Unit Mathematics exam
- Geometric Applications of the Derivative
- Quadratic Functions
- Locus and the Parabola
- Circle Geometry
- 3 Dimensional Trigonometry
- Trigonometric Sums and Difference and the Angle Between Two Lines
- Harder Inequalities
- Division of an Interval in a Given Ratio

QUESTION 1 (13 Marks)**[START A NEW BOOKLET]****Marks**

- (a) Show that if $f(x) = 2x(1+4x)^5$ then $f'(x) = 2(24x+1)(1+4x)^4$ **2**
- (b) Solve $\frac{7}{x-2} \geq 3$ **3**
- (c) Factorise $9a^3 - 72b^3$ **2**
- (d) Solve $\tan^2(2\theta) = 3$ for $-180^\circ \leq \theta \leq 180^\circ$ **3**
- (e) Find the equation of the locus of a point $P(x,y)$ that moves such that it is always equidistant from $(3,1)$ and $(-1,-2)$ **3**

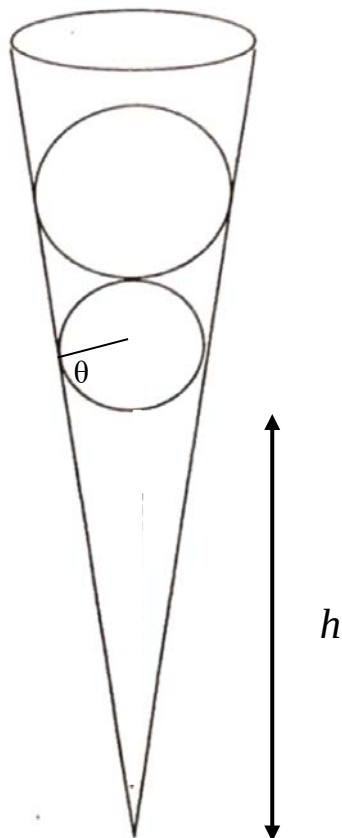
End of Question 1

- (a) For the curve $y = x^3 - 3x + 3$,
- (i) Find the coordinates of any stationary points and determine their nature. 3
 - (ii) Find any points of inflexion. 2
 - (iii) Sketch the curve showing the above information.
Your graph should take about $\frac{1}{3}$ of the page. 2
 - (iv) Find the equation of the tangent at the point of inflexion 2
- (b) Find the coordinates of the focus and the equation of the directrix of $y^2 = 8x$ 2
- (c) Sketch the gradient function of the following graph. Label any x intercepts on your graph. 2



End of Question 2

- (a) If α and β are the zeros of the function $y = x^2 + 2x - 1$, find the value of $\alpha^2 + \beta^2$ 3
- (b) Find the obtuse angle between the lines $4x - y + 6 = 0$ and $x + 3y - 7 = 0$.
Give your answer to the nearest minute. 3
- (c) (i) Give the expression for $\tan(2x)$ in terms of $\tan x$ 1
- (ii) Hence show if $f(x) = x \cot x$ then $f(2x) = (1 - \tan^2 x) f(x)$ 3
- (d)



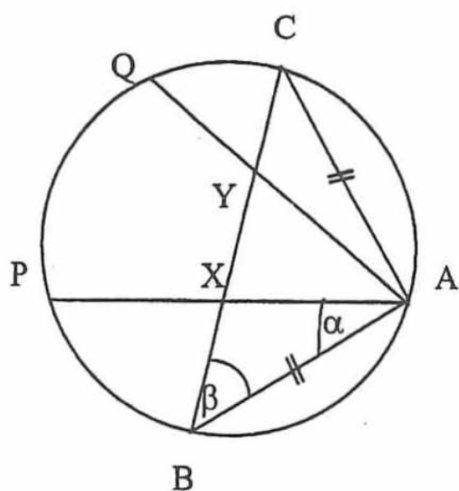
Two marbles of radius 3cm and 5cm are placed in an inverted cone so that the marbles touch each other and the sides of the cone.

- (i) Find θ , the angle the smaller marble makes with the side of the cone at the point where they touch. Give a reason for your answer. 1
- (ii) Using similar triangles, find the distance h from the vertex of the cone to the smallest marble. 2

End of Question 3

- (a) Find the domain and range of $y = \sqrt{x^2 - 16}$ 2
- (b) Point A has coordinates $(-3, 1)$ and point B has coordinates $(b, -4)$.
The point $P(11, -8)$ divides the interval AB externally in the ratio $9:4$. Find b . 2
- (c) Two houses, P and Q , lie in the same plane as S . S is the foot of a hill RS .
The height of the hill is 200m and the angle of elevation from P to the top of the hill is 14° .
 P and Q subtend an angle of 75° at S and Q is 500m from S .
- (i) Draw a diagram to show the above information. 1
- (ii) Find the distance between the two houses, to the nearest metre. 3

(d)



Let $ABPQC$ be a circle with $AB = AC$.

AP intersects BC at X and AQ intersects BC at Y . Let $\angle PAB = \alpha$ and $\angle ABC = \beta$.

Trace or draw the diagram in your answer booklet.

- (i) Explain why $\angle AXC = \alpha + \beta$ 1
- (ii) Explain why $\angle PQB = \alpha$ 1
- (iii) Prove $XYQP$ is a cyclic quadrilateral 2

End of Question 4

QUESTION 5 (12 Marks)**[START A NEW BOOKLET]****Marks**

- (a) Find the values of m for which $-x^2 + (m-2)x - 25$ has no real roots. **3**
- (b) A piece of wire 84cm long is to be cut into two pieces.
One piece is used to form a square and the other a rectangle 3 times as long as it is wide.
- (i) Let x be the side length of the square and y be the width of the rectangle.
Show the sum of the areas of the square and rectangle is $A = 7(y^2 - 12y + 63)$ **3**
- (ii) Find the lengths of both pieces if the sum of the areas is to be a minimum. **3**
- (c) (i) Simplify $(\cos x + \sin x)(\cos x - \sin x)$ **1**
- (ii) Hence, or otherwise, prove $\frac{\cos x + \sin x}{\cos x - \sin x} = \frac{\sin 2x + 1}{\cos 2x}$ **2**

End of Question 5

QUESTION 6 (12 Marks)**[START A NEW BOOKLET]****Marks**

- (a) Consider the function $f(x) = \frac{2x^2}{1-x^2}$
- (i) Show $f(x)$ is an even function. **1**
- (ii) Find any vertical asymptotes of $f(x)$. **1**
- (iii) Find the turning point and determine its nature. **3**
- (iv) Write the following sentence in your answer booklet and fill in the blank:
As x approaches $\pm\infty$, y approaches _____ **1**
- (v) Sketch $f(x)$ **2**
- (b) (i) Sketch $y = \frac{5}{|x-1|}$ **2**
- (ii) Hence, or otherwise, find where $y < 3$ **2**

End of Question 6
End of Paper

Year 11 Ext 1 Yearly Exam T3 2017 Student Solutions

Q1 a) $f'(x) = 2x \times 5 \times 4(1+4x)^4 + 2(1+4x)^5$
 $= 40x(1+4x)^4 + 2(1+4x)^5$
 $= 2(1+4x)^4 [20x + 1 + 4x]$
 $= 2(24x+1)(1+4x)^4$

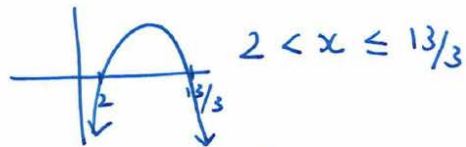
b) $x \neq 2 \quad \frac{7}{x-2} - 3 \geq 0$

$\frac{7}{x-2} - \frac{3(x-2)}{x-2} \geq 0$

$\frac{13-3x}{x-2} \geq 0$

$(x-2)^2 \frac{(13-3x)}{(x-2)} \geq 0$

$(x-2)(13-3x) \geq 0$



c) $9(a-2b)(a^2+2ab+4b^2)$

d) $\tan 2\theta = \pm\sqrt{3} \quad -360^\circ \leq 2\theta \leq 360^\circ$

$2\theta = \pm 60^\circ, \pm 120^\circ, \pm 240^\circ, \pm 300^\circ$

$\theta = \pm 30^\circ, \pm 60^\circ, \pm 120^\circ, \pm 150^\circ$

e) $P(x,y)$ let $A(3,1)$ $B(-1,-2)$

Now $PA^2 = PB^2$

$(x-3)^2 + (y-1)^2 = (x+1)^2 + (y+2)^2$

$x^2 - 6x + 9 + y^2 - 2y + 1 = x^2 + 2x + 1 + y^2 + 4y + 4$

$-8x - 6y + 5 = 0$

$8x + 6y - 5 = 0$

Q2 a) (i) $y' = 3x^2 - 3$

$y'' = 6x$ for stat pts $y' = 0$

$0 = 3x^2 - 3$

$0 = x^2 - 1$

$0 = (x+1)(x-1)$

when $x = 1$ $y'' = 6 > 0$

$\therefore (1, 1)$ min. t.p.

when $x = -1$ $y'' = -6 < 0$

$\therefore (-1, 5)$ max. t.p.

(ii) for inflexions $y'' = 0$

$6x = 0$

$x = 0$

$y = 3$

$\therefore (0, 3)$ possible POI

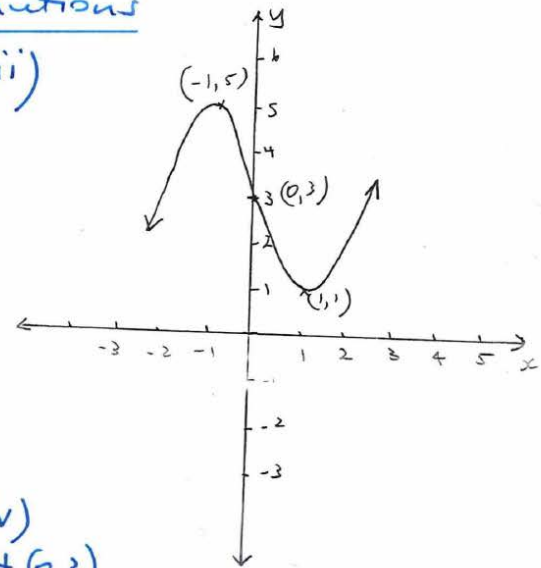
x	-1	0	1
y''	-6	0	6

$\therefore$ change in concavity

$\therefore (0, 3)$ POI.

①

(iii)



(iv)

at $(0, 3)$

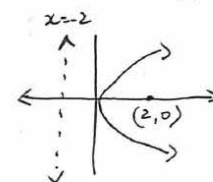
$y' = 3x^2 - 3$

$m = -3$

$y - 3 = -3(x - 0)$

$y = -3x + 3$ or $3x + y - 3 = 0$

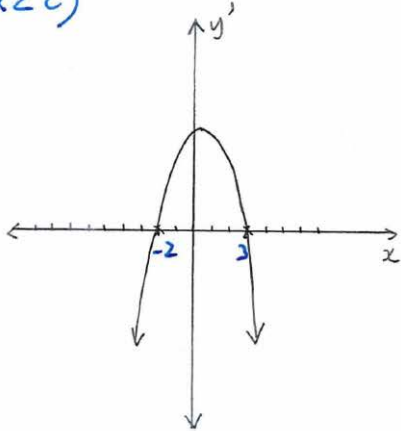
b) $y^2 = 8x$ vertex $(0, 0)$
 $a = 2$



focus $(2, 0)$

directrix $x = -2$

Q2c)



Q3 a) $\alpha + \beta = -2$ $\alpha\beta = -1$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (-2)^2 - 2(-1) = 6$$

b) $4x - y + 6 = 0$ $y = 4x + 6$
 $m_1 = 4$

$x + 3y - 7 = 0$ $y = -\frac{1}{3}x + \frac{7}{3}$
 $m_2 = -\frac{1}{3}$

$\tan \theta = \left| \frac{4 + \frac{1}{3}}{1 - 4 \cdot \frac{1}{3}} \right| = 13$
 $\theta = 85^\circ 36'$

obtuse $\theta = 180 - 85^\circ 36'$
 $= 94^\circ 23' 55.34''$
 $= 94^\circ 24' \text{ (nearest min)}$

c) (i) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

c) (ii) $f(2x) = 2x \cot 2x$

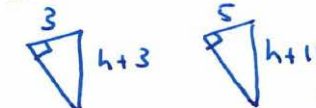
$$= \frac{2x}{\tan 2x}$$

$$= 2x \times \frac{(1 - \tan^2 x)}{2 \tan x}$$

$$= x \cot x (1 - \tan^2 x)$$

$$= (1 - \tan^2 x) f(x).$$

d) (i) $\theta = 90^\circ$ the angle between the radius and tangent to a circle is $\frac{\pi}{2}$.

(ii)  $\frac{3}{5} = \frac{h+3}{h+11}$
 $3h + 33 = 5h + 15$
 $h = 9$

Q4 a) $x^2 - 16 \geq 0$ $(x-4)(x+4) \geq 0$

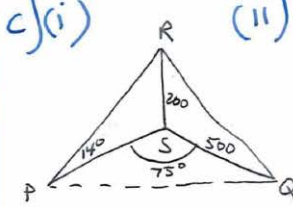
Domain
 $x \geq 4$
 $x \leq -4$

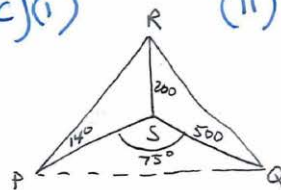


Range $y \geq 0$

b) $11, -8 = \left(\frac{9b+12}{5}, \frac{-36-4}{5} \right)$

$$\frac{9b+12}{5} = 11 \quad b = 43/9$$

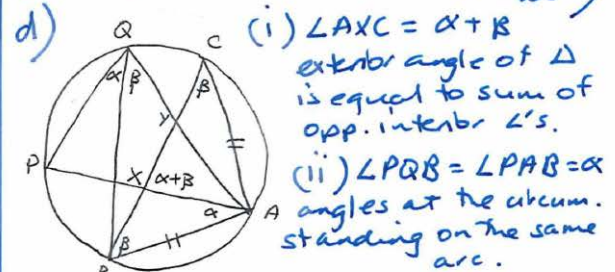
c) (i)  (ii) $PS = \frac{200}{\tan 14^\circ}$



(2)

$$PQ^2 = \left(\frac{200}{\tan 14^\circ} \right)^2 + 500^2 - 2 \left(\frac{200}{\tan 14^\circ} \right) \times 500 \cos 75^\circ$$

$$PQ = 828.16 = 828 \text{ m (nearest m)}$$



(iii) In ΔABC $\angle ACB = \beta$ (base angles in isos. Δ)
 Now $\angle BQA = \beta$ (angles at the circum. standing on the same arc.)

From (i) $\angle AXB = \alpha + \beta$
 $\therefore \angle YXP = 180 - (\alpha + \beta)$ (straight line)

$\angle PQA = \alpha + \beta$ on diagram
 Now $\angle PQA + \angle YXP = 180^\circ$
 $(\alpha + \beta) + (180 - \alpha + \beta) = 180^\circ$
 $\therefore XYQP$ is a cyclic quadrilateral since opp $\angle$'s supplementary.

Q5 a) $\Delta = (m-2)^2 - 4 \times -1 \times -25$
 $= m^2 - 4m - 96$

for no real roots $\Delta < 0$

$$m^2 - 4m - 96 < 0$$

$$(m-12)(m+8) < 0$$

$$-8 < m < 12$$



5b) 

$$A = x^2$$

$$P = 4x$$

$$A = 3y^2$$

$$P = 8y$$

$$4x + 8y = 84$$

$$x = 21 - 2y$$

$$(i) \text{ Sum Areas} = x^2 + 3y^2$$

$$= (21 - 2y)^2 + 3y^2$$

$$= 7y^2 - 84y + 441$$

$$= 7(y^2 - 12y + 63)$$

$$(ii) A' = 14y - 84 \text{ for min } A' = 0$$

$$0 = 14y - 84$$

$$y = 6$$

$$A'' = 14 > 0 \therefore y = 6 \text{ is a min}$$

$$x = 21 - 2(6)$$

$$x = 9$$

$$\text{length } P_1 = 4x = 4(9) = 36\text{cm}$$

$$\text{length } P_2 = 8y = 8(6) = 48\text{cm}$$

$$\text{check } 36 + 48 = 84\text{cm}$$

$$5c) (i) \cos^2 x - \sin^2 x$$

$$(ii) \text{ RHS} = \frac{2 \sin x \cos x + 1}{\cos^2 x - \sin^2 x}$$

$$\text{LHS} = \frac{\cos x + \sin x}{\cos x - \sin x} \times \frac{\cos x + \sin x}{\cos x + \sin x}$$

$$= \frac{\cos^2 x + 2 \sin x \cos x + \sin^2 x}{\cos^2 x - \sin^2 x}$$

$$= \frac{1 + 2 \sin x \cos x}{\cos^2 x - \sin^2 x} = \text{RHS}$$

$$Q6 (i) f(-x) = \frac{2(-x)^2}{1 - (-x)^2} = \frac{2x^2}{1 - x^2} = f(x)$$

$\therefore$ even function

$$(ii) 1 - x^2 = 0$$

$$(1 - x)(1 + x) = 0$$

$\therefore x = 1$ or $x = -1$ are vertical asymptotes

$$(iii) f'(x) = \frac{(1 - x^2)(4x) - 2x^2(-2x)}{(1 - x^2)^2}$$

$$= \frac{4x}{(1 - x^2)^2} \text{ for t.p. } f'(x) = 0$$

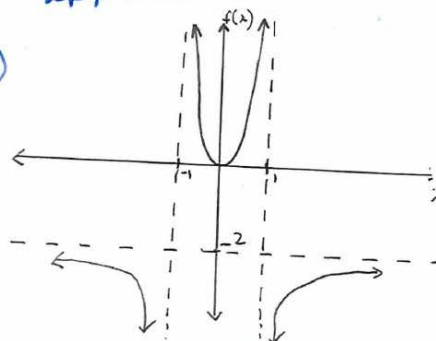
$$0 = \frac{4x}{1 - x^2} \therefore x = 0, y = 0$$

$\therefore (0, 0)$ min t.p.

x	-1.2	0	1.2
f(x)	-3.5	0	3.5
shape	\	/	/

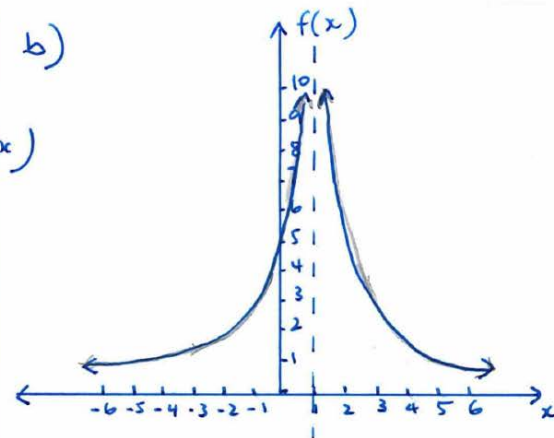
(iv) as x approaches $\pm \infty$, y approaches -2

(v)



(3)

b)



$$(ii) 3 = \frac{5}{x - 1}$$

$$3x - 3 = 5$$

$$x = 8/3$$

$$3 = \frac{-5}{x - 1}$$

$$3(x - 1) = -5$$

$$x = -\frac{2}{3}$$

or draw $y < 3$ on graph above

$$\therefore x < -\frac{2}{3} \text{ or } x > \frac{8}{3}$$